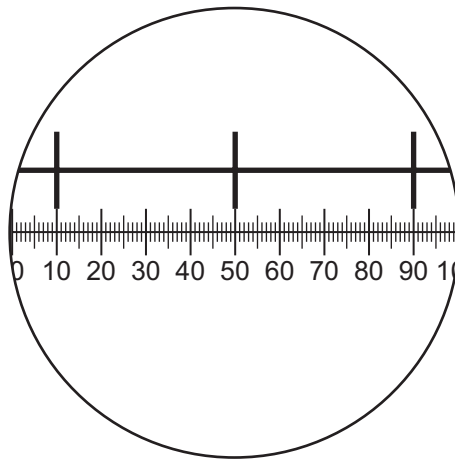


- 1 The diagram shows a stage micrometer, with divisions 0.10 mm apart, viewed through an eyepiece containing a graticule.



The area of the field of view of the microscope can be calculated using this formula.

$$\text{area} = \pi r^2$$

A student calculated the area of the field of view of the microscope using the information provided and a value for π of 3.142.

Which answer has been rounded correctly to an appropriate number of significant figures?

- A 0.04909 mm²
- B 5×10^{-2} mm²
- C 4.909×10^4 μm^2
- D 4.91×10^4 μm^2